CSCI 2406: Topic 01

Computer Architecture & Organization Fundamentals

Abstraction Layers, Architecture vs. Organization, Stored-Program Concept, and Memory Models in AArch64

Computer Organization & Architecture | Topic 1

Course Information & Administration

Instructor Contact

- **Email**: `jburns@ada.edu.az`
- **Office**: B317
- **Office Hours**: Thursdays 11:00–15:00 (or by appointment)
- *Note*: Avoid Blackboard messages for urgent matters.

Classroom Expectations

- **The Golden Rule**: Once class commences and the instructor speaks, no outside conversations are permitted.
- **Authentic Work**: Emphasize the *why* alongside the *what* and *how*. Do not pass off AI-generated output as your own.

Tools, Emulation, and Assessments

Software & Emulation Stack

- **C & ARM64 Assembly**: Core languages utilized for architecture programming.
- **QEMU**: Full-system and user-space emulator for AArch64 virtual execution.
- **Compiler Explorer**: Live ARM64 assembly inspection from C source.

Online Programming Resources

- **WebAssembliss**: Browser-based ARM64 Linux assembly environment: `https://web.assembliss.app/editor/arm64_linux/`
- **CPUlator**: Interactive web-based ARM32 / ARMv7 system simulator: `https://cpulator.01xz.net/?sys=arm`

Topic 1 Outline

Foundational Focus

Understanding the bridge connecting high-level algorithms to the physical silicon execution substrate in modern 64-bit ARM systems.

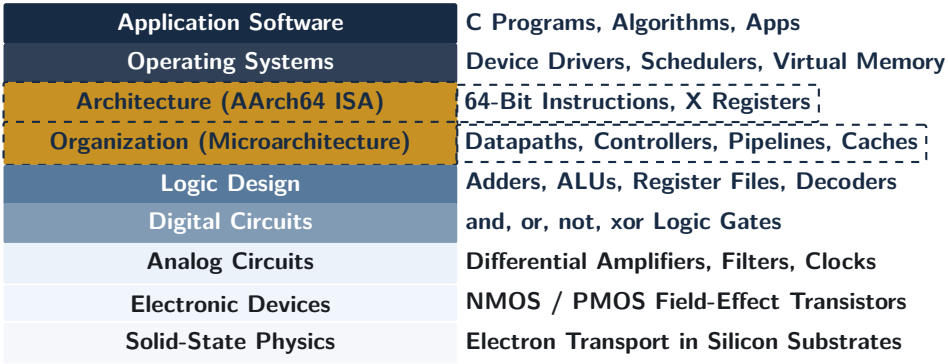
- **Section 1**: The Full Stack of Abstraction Layers
- **Section 2**: Computer Architecture versus Computer Organization
- **Section 3**: Instruction Set Architecture (ISA) versus Microarchitecture
- **Section 4**: The Stored-Program Concept & Execution Cycle
- **Section 5**: Memory Models: von Neumann, Harvard, and Modified Harvard
- **Section 6**: Summary and Interactive Self-Test Review

Section 01

The Full Stack of Abstraction Layers

The 9 Levels of Computer Abstraction

Computers manage immense complexity by enforcing clean hierarchical interfaces:



Where Software Meets Hardware

CSCI 2406 Focus

Our course focuses on the critical boundary that translates logical intent into physical electrical actions: computer architecture and organization.

Above the Boundary (Architecture / Software)

- Applications compiled into machine binaries.
- Operating systems managing hardware through architectural exception levels.
- Completely portable across any hardware implementing the identical AArch64 ISA.

Below the Boundary (Organization / Hardware)

- Organization and microarchitecture (datapaths and state controllers).
- Digital circuits and gate-level logic.
- Silicon fabrication nodes and thermal design power (TDP) constraints.

Section 02

Computer Architecture vs. Computer Organization

What is Computer Architecture?

Computer Architecture defines the **externally visible capabilities and behavior** of the machine—the ***what***.

- The programmer-visible specification of the computer system.
- Establishes what the software is allowed to execute and what state changes occur.
- Determines the binary interface targeted by compilers and assembly programmers.

Architectural Specifications Include:

- The complete **Instruction Set Architecture (ISA)** vocabulary (AArch64).
- Programmer-visible registers (`x0`–`x30`, stack pointer `sp`, program counter `pc`, status flags).
- Instruction encodings, data formats (byte, halfword, word, doubleword), and addressing modes.

What is Computer Organization?

Computer Organization describes the **internal hardware implementation** of the architecture—the ***how***.

- The physical arrangement of operational units and their logical interconnections.
- Defines how the machine physically performs the architectural operation.
- Completely transparent (invisible) to the programmer and machine software.

Organizational Structures Include:

- Datapaths, control units, ALUs, and multiplexer routing.
- Pipeline staging, branch prediction logic, and cache hierarchies (L1/L2/L3).
- Internal bus interfaces, clocking frequencies, and memory controller arbitration.

Architecture vs. Organization: Direct Contrast

Computer Architecture (*What*)	**Computer Organization (*How*)**
Programmer / Compiler view	Hardware engineer / Silicon designer view
Functional contract and behavior	Physical implementation and datapath
Changes slowly across generations	Evolves rapidly with silicon manufacturing nodes
Example: Does a 64-bit multiply exist?	Example: Is multiplication implemented via a hardware multiplier array or Booth recoder?
Specifies visible registers & flags	Specifies pipeline stages, ALU internal circuits
Software target (C, ARM64 assembly)	Hardware target (Verilog, VHDL, RTL)

Architectural Stability and Hardware Evolution

The Benefit of a Stable Architecture

Preserving the instruction set architecture allows software investments to persist indefinitely while hardware design is continuously overhauled to boost performance.

Architectural Invariance

- The **AArch64 ISA** defines standard instructions like `add`, `ldr`, and `cbz`.
- An ARM64 binary compiled for early AArch64 systems executes seamlessly on modern Apple M-series or AWS Graviton CPUs.

Organizational Evolution

- **Early Cores**: In-order or narrower out-of-order execution pipelines.
- **Modern Cores**: Wide-issue out-of-order engines, deep multi-level caches, and advanced execution units.

Section 03

ISA versus Microarchitecture

The Instruction Set Architecture (ISA)

The **ISA** is the complete vocabulary of machine instructions supported by the hardware.

- All software running on a given processor must be expressed using its specific ISA.
- Different programs use different subsets, but all conform to the common ISA dictionary.

The Architectural Contract: `add x1, x2, x3`

The ISA guarantees a single semantic state change:

$$\text{Register } x_1 \leftarrow \text{Register } x_2 + \text{Register } x_3$$

- The architecture defines *what* state update occurs upon completion.
- It does not state *how* data moves across physical buses or wires inside the CPU.

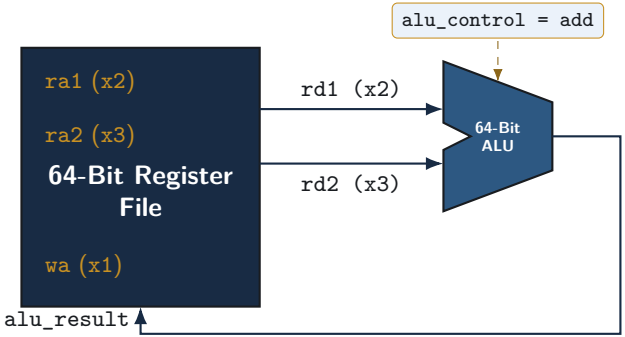
Microarchitecture: Executing `add x1, x2, x3`

Register Transfer Steps for `add x1, x2, x3`

1. `ra1 <- x2; ra2 <- x3` *(Assert register read addresses)*
2. `rd1 <- RegisterFile[ra1]; rd2 <- RegisterFile[ra2]` *(Read 64-bit operand values)*
3. `alu_control <- add` *(Configure ALU operation multiplexers)*
4. `alu_result <- rd1 + rd2` *(Execute 64-bit addition)*
5. `reg_write <- 1` *(Assert register file write enable)*
6. `RegisterFile[x1] <- alu_result` *(Commit result to destination register)*

- **ISA View**: Add 64-bit values in `x2` and `x3`; store sum into `x1`.
- **Microarchitecture View**: Sequence control signals, bus transfers, and ALU operations.

Visualizing the Microarchitectural Datapath



- Microarchitectures differ by doing this sequentially in multiple cycles or parallelized within single/pipelined cycles.

One ISA — Multiple Hardware Realizations

Microarchitecture	Datapath Style	Design Objective
Single-Cycle	Every instruction takes 1 long clock cycle	Simple control; slow clock rate
Multi-Cycle	Instructions take variable number of short cycles	Low hardware cost; reuse functional units
Pipelined (Classic 5-Stage)	Overlap fetch, decode, execute, memory, writeback	Higher instruction throughput (1 inst/cycle ideal)
Superscalar Out-of-Order	Multiple pipelines, dynamic instruction scheduling	Maximum performance for desktops/servers

Key Principle

All four microarchitectures execute the identical compiled AArch64 binary accurately.

Section 04

The Stored-Program Concept

The Stored-Program Concept

Foundational Definition

Program instructions are encoded as binary numbers and stored in the exact same addressable memory subsystem alongside standard runtime data.

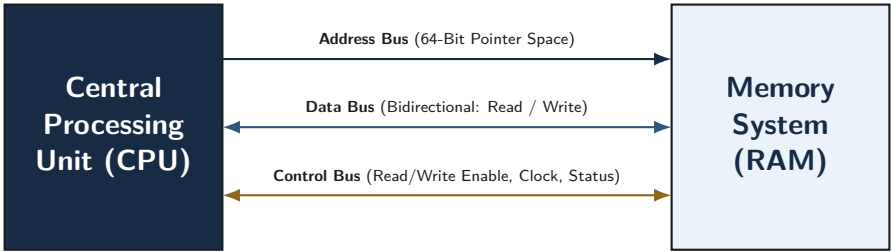
Capabilities of a General-Purpose Computer:

- **Store Information**: Holds data values and program sequences electronically.
- **Manipulate Information**: Performs mathematical and logical transformations via the ALU.
- **Make Decisions**: Alters control flow dynamically based on evaluated conditions (e.g., conditional branches).

Core Advantage

Reprogramming does not require physical rewiring; it simply requires loading new numbers into memory.

The CPU–Memory Interface: Three Fundamental Buses



- **Address Bus**: Selects the specific memory cell location using 64-bit addresses.
- **Data Bus**: Transfers the numerical word to or from the selected cell.
- **Control Bus**: Dictates the direction of transfer (read vs. write) and timing signals.

Memory as an Array of Numbered Cells

How Memory is Structured:

- Memory is viewed logically as a contiguous 1-D array of cells.
- Each cell has a unique numerical **Address** and stores a **Value**.
- Address 4 holds value 3.
- An address is simply an index; the data is the payload.

Address	Stored Value
0	7
1	3
2	15
3	20
4	**3** ← Cell content
5	7
6	3

Context Determines Meaning

Bit Invariance Principle

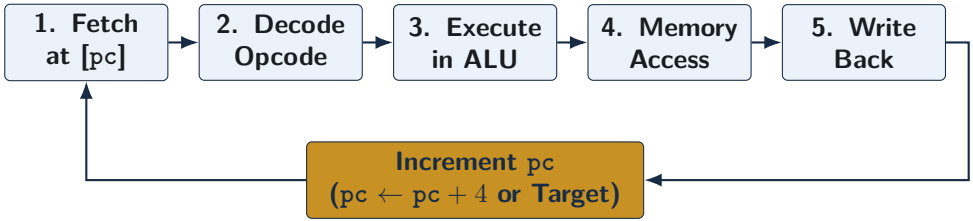
A pattern of bits residing in a memory cell possesses no intrinsic, permanent identity. Its meaning is determined entirely by **how the CPU accesses and interprets it**.

Consider a 32-bit word stored in memory at location `0x0x0000000000400000`:

- If the **program counter (`pc`)** points here $\rightarrow$ the CPU fetches and decodes it as an **AArch64 instruction**.
- If an **arithmetic load instruction** reads this address $\rightarrow$ the CPU treats it as a **signed / unsigned integer**.
- If a **pointer variable** holds this address $\rightarrow$ it is treated as a **64-bit memory reference**.
- If passed to a **display routine** $\rightarrow$ it is rendered as **ASCII/UTF-8 text characters**.

The Classical Machine Instruction Lifecycle

The processor continuously steps through code using the **program counter (pc)**:



- Because code is in memory, programs can be loaded, copied, relocated, or modified dynamically.

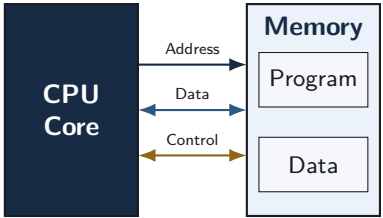
Section 05

von Neumann vs. Harvard Architectures

The von Neumann Architecture

Core Characteristics:

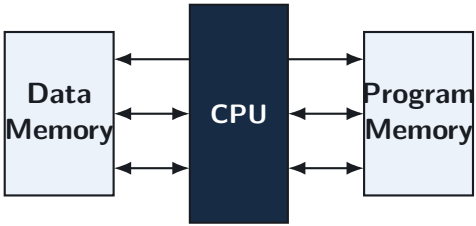
- **Unified Memory**: A single address space containing both instructions and data.
- **Shared Bus System**: A single set of address, data, and control lines is shared.
- **Operation**: The CPU fetches an instruction, then subsequently accesses data using the same bus.
- Directly realizes the stored-program concept in its purest form.



The Harvard Architecture

Core Characteristics:

- **Separated Memories**: Physically independent instruction and data memory modules.
- **Independent Buses**: Dedicated address, data, and control lines for each memory bank.
- **Parallel Operation**: CPU can fetch an instruction and read/write data operands **simultaneously**.
- Instruction and data words can have differing bit-widths and storage technologies.



Structural Separation: The Key Distinction

The Fundamental Question

The decisive architectural issue is not simply where bytes are stored. It is **whether instruction access and data access are structurally separated**.

Single Shared Bus (vN)

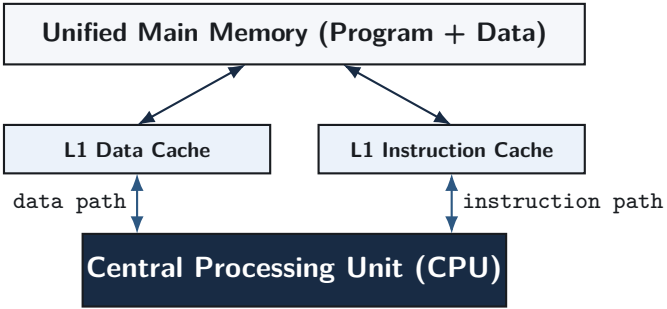
- Sequential access creates bus contention (**von Neumann bottleneck**).
- Lower hardware routing cost and flexible memory budgeting.

Dedicated Dual Buses (Harvard)

- Concurrent instruction fetch and operand read/write eliminates structural stalls.
- Higher pin counts and rigid memory partition limits.

The Modern Standard: Modified Harvard Architecture

Modern RISC systems (e.g., Apple M-series, ARM Cortex-A64) merge both concepts:



- **Internally**: Separate L1 caches yield Harvard pipeline throughput.
- **Externally**: Unified RAM preserves von Neumann software simplicity.

Comparison Summary: Memory Topologies

Model	Bus & Memory Structure	Primary Modern Use Case
von Neumann	Single shared memory and bus for instructions and data	Low-cost microcontrollers, simple embedded state controllers
Harvard	Physically separate memories and buses for code vs. data	Digital Signal Processors (DSPs), ATmega/AVR Arduino boards
Modified Harvard	Split L1 caches inside CPU; unified main memory outside	Modern high-performance RISC / CISC (ARM Cortex, Apple Silicon, x86-64)

Section 06

Summary & Self-Test

Topic Summary

- **Architecture vs. Organization**: Architecture is the programmer-visible specification (the *what*); Organization is the internal hardware implementation (the *how*).
- **ISA vs. Microarchitecture**: ISA is the instruction vocabulary contract; Microarchitecture specifies the datapath and control sequencing.
- **Stored-Program Concept**: Instructions are stored as binary numbers in memory alongside data; meaning is determined by how the CPU accesses them.
- **von Neumann Architecture**: Shares a single memory and bus system for both instructions and data.
- **Harvard Architecture**: Uses separate memory modules and dedicated buses for concurrent instruction and data access.
- **Modified Harvard**: Blends separate L1 caches near the CPU core with unified main memory.

Self-Test: Questions 1 & 2

Question 1: Architecture vs. Organization

What is the difference between computer architecture and computer organization? Provide two specific features belonging to each domain.

Question 2: ISA vs. Microarchitecture

For the AArch64 instruction `add x1, x2, x3`, what does the ISA specify, and what internal actions belong to the microarchitecture?

Question 3: The Stored-Program Concept

What is the stored-program concept, and why was it essential for creating general-purpose computing machines?

Self-Test: Solutions 1 & 2

Solution 1

Architecture is the software-visible specification (e.g., 64-bit register count, ISA opcodes).
Organization is the hardware implementation (e.g., pipeline depth, cache associativity).

Solution 2

- **ISA Specifies**: Add contents of `x2` to `x3` and place the 64-bit result in `x1`.
- **Microarchitecture Executes**: Read registers `x2` and `x3` via port addresses `ra1`/`ra2`, configure ALU control to `add`, assert write-enable (`reg_write=1`), and commit to `x1`.

Self-Test: Questions 4 & 5

Question 4: Memory Model Topologies

What is the primary structural difference between von Neumann and Harvard architectures, and why can Harvard machines achieve higher cycle efficiency?

Topic 1 Self-Test Question 5: Modified Harvard Design

Explain what a Modified Harvard architecture is. How does Apple Silicon or ARM implement this concept in hardware?

Self-Test: Solutions 4 & 5

Solution 4

von Neumann uses a single unified bus and memory, causing serial access contention. Harvard uses independent buses and memories, allowing simultaneous instruction fetch and data operand load/store in the same clock cycle.

Solution 5

A Modified Harvard architecture provides separate L1 instruction and data caches directly adjacent to the CPU core to eliminate pipeline stalls, while utilizing a unified L2/L3 cache and main memory system to maintain a unified address space.

Wrap-Up and Next Steps

Transition to Topic 2

Building upon the foundational abstraction layers, system organization, and memory models, the next module examines:

- **RISC Design Principles**: Exploring the philosophy of Reduced Instruction Set Computers, simpler control circuitry, and high energy efficiency.
- **The Load/Store Paradigm**: Examining how RISC architectures strictly isolate memory access from register-based computation.
- **CISC versus RISC Trade-Offs**: Contrasting variable-length instructions and direct memory arithmetic with fixed 32-bit execution models.
- **AArch64 Architectural Overview**: Analyzing the modern 64-bit ARM instruction encoding model and dual register views (`W` vs. `X`).
- **Programmer-Visible State**: Surveying general-purpose registers (`x0`–`x30`), stack pointers, link registers, and `PSTATE` condition flags.